Maximal Volume Ideal Polyhedra and the Arithmetic Angle Phenomenon

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Abstract

We present a software suite for the analysis and optimization of ideal convex polyhedra in hyperbolic 3-space $\mathbb{H}^3$. Using Rivin's variational characterization of ideal polyhedra, we develop efficient algorithms for checking combinatorial realizability and finding volume-maximizing configurations. Our systematic computational study reveals two striking phenomena: (1) maximal volume ideal polyhedra consistently exhibit dihedral angles that are rational multiples of π—a property with no obvious explanation from the optimization formulation; and (2) the distribution of volumes for random configurations is well-approximated by a Beta distribution, with mean normalized volume converging to approximately $\ln 2 \approx 0.69$ as the vertex count increases. We provide complete data for small vertex counts, including vertex positions, triangulations, and verified rational angle structures. Extending to $\mathbb{H}^4$ and $\mathbb{H}^6$, we discover a *combinatorial invariance* phenomenon: the f-vector of the volume-maximizing ideal polytope with N vertices on S^{d-1} is the same whether volume is measured in the Euclidean or hyperbolic sense, holding exactly for $N \lesssim 5d$ (tested up to $d = 3$, $N \leq 50$; $d = 4$, $N \leq 40$; $d = 5$, $N \leq 20$; $d = 6$, $N \leq 30$). At larger N, small deviations appear (≤ 3 facets), with the hyperbolic optimizer consistently finding fewer facets. CMA-ES (Covariance Matrix Adaptation) provides Euclidean warm starts that are within 0.0–0.4% of the hyperbolic optimum, confirming the near-equivalence of the two objectives near their shared maximum. An interactive implementation is publicly available.

Keywords: ideal polyhedra, hyperbolic volume, Lobachevsky function, dilogarithm, combinatorial invariance, Frobenius groups, vertex-transitive polytopes, CMA-ES, Schläfli formula

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Contents

1 Introduction **3**

2 Background and Notation **4**
2.1 Ideal Polyhedra in $\mathbb{H}^3$ 4
2.2 Rivin's Variational Characterization 4
2.3 The Lobachevsky Function and Hyperbolic Volume 5
2.4 The Penner-Rivin Algorithm 6

3 Methods **6**
3.1 Dimension 3: Angle-Based Optimization 7
3.1.1 Realizability testing 7
3.1.2 Volume optimization in angle coordinates 7
3.1.3 Direct vertex optimization via Bloch–Wigner 7
3.1.4 Rational angle detection 8
3.2 Dimensions $d \geq 4$: Vertex-Based Optimization 8
3.2.1 Stage 1: Euclidean warm start via CMA-ES 8
3.2.2 Stage 2: Hyperbolic polish 8
3.2.3 Stage 3: Triangulation management 9
3.2.4 Structured initialization: a small-N phenomenon . . . 10
3.3 Symmetry Analysis 11
3.4 Hessian Analysis 11

4 Results **11**
4.1 Maximal Volume Configurations 11
4.2 The Rational Angle Phenomenon 13
4.3 Connection to Arithmetic Groups 16
4.4 Volume Distribution of Random Configurations 17

5 Visualizations **19**

6 Software **19**

7 Higher Dimensions: Ideal Polytopes in $\mathbb{H}^4$ **20**
7.1 Setup 20
7.2 Combinatorial Invariance of Optimal Polytopes 20
7.3 Cross-Metric Divergence 22
7.4 Correlation Between Euclidean and Hyperbolic Volumes . . . 23
7.5 Non-Isolated Maxima: The Volume Plateau 24
7.6 Higher Dimensions 25

7.7 Ideal Polytopes in $\mathbb{H}^5$ 26
7.8 Ideal Polytopes in $\mathbb{H}^6$ 27
7.9 Identification of Optimal Polytopes: Joins of Regular Polygons 27
7.10 Symmetry Analysis via Douady–Earle Centering 29
7.10.1 The Frobenius construction 31
7.10.2 Volume ridges 31
7.11 Isoperimetric Ratio 32

8 LLM Benchmarks: A Pointer **33**
8.1 The reconstruction problem and rigidity 34
8.2 Volume lower bound and our records 34

9 Open Questions **35**

10 Application: High-Girth Cubic Graphs **37**
10.1 Method 37
10.2 Results 38

1 Introduction

Ideal polyhedra in hyperbolic 3-space $\mathbb{H}^3$ occupy a central position in low-dimensional topology and geometry. Their study connects to the geometry of hyperbolic 3-manifolds, Teichmüller theory, and the representation theory of surface groups. The volume of hyperbolic 3-manifolds, a topological invariant, can often be computed by decomposing the manifold into ideal tetrahedra—the simplest ideal polyhedra.

A natural question arises: among all ideal polyhedra with a given number of vertices, which has the largest volume? This optimization problem, while analytically straightforward (it reduces to maximizing a concave function over a convex polytope), reveals surprising structure. The maximal configurations exhibit a form of "arithmeticity"—their dihedral angles are consistently rational multiples of π. Moreover, the distribution of volumes for random configurations follows a Beta distribution, with parameters scaling linearly in the vertex count and mean normalized volume converging to approximately $\ln 2$.

In this paper, we present a comprehensive software suite for:

(i) Checking whether a given triangulation is realizable as the boundary of an ideal convex polyhedron

(ii) Finding optimal (volume-maximizing) geometric realizations of realizable triangulations

(iii) Computing the Fuchsian group associated with the surface of the polyhedron

(iv) Visualizing polyhedra in multiple models of hyperbolic space

Our computational experiments reveal a surprising phenomenon: when optimizing for maximal volume, the resulting polyhedra consistently have dihedral angles that are *exact rational multiples of* π. This arithmetic structure emerges from a purely analytic optimization problem with no built-in number-theoretic constraints.

2 Background and Notation

2.1 Ideal Polyhedra in $\mathbb{H}^3$

An *ideal polyhedron* in hyperbolic 3-space $\mathbb{H}^3$ is a convex polyhedron whose vertices lie on the boundary at infinity $\partial_\infty \mathbb{H}^3 \cong \mathbb{CP}^1$. Despite having vertices "at infinity," such polyhedra have finite volume—a consequence of the exponential shrinking of hyperbolic space toward the boundary.

The combinatorial structure of an ideal polyhedron is encoded by a triangulation of the 2-sphere: the *Delaunay triangulation* of the ideal vertices as seen from the center of mass of the polyhedron. Conversely, not every triangulation of the sphere arises from an ideal polyhedron—realizability imposes constraints on the possible angle structures, as characterized by Rivin's theorem below.

We work primarily in the upper half-space model, where $\mathbb{H}^3 = \{(x, y, z) \in \mathbb{R}^3 : z > 0\}$ with metric $ds^2 = (dx^2 + dy^2 + dz^2)/z^2$. The boundary $\partial_\infty \mathbb{H}^3$ is the extended complex plane $\mathbb{C} \cup \{\infty\}$, and ideal vertices are specified as points in $\mathbb{CP}^1$.

2.2 Rivin's Variational Characterization

The foundational result for our approach is Rivin's characterization of ideal polyhedra:

Theorem 1 (Rivin [1, 2])**.** *A triangulation $\mathcal{T}$ of the sphere is realizable as the boundary of a convex ideal polyhedron in $\mathbb{H}^3$ if and only if there exist positive real numbers $\theta_c > 0$ assigned to each corner c of each triangle such that:*

(i) ***Triangle constraint:*** *For each triangle T with corners c_1, c_2, c_3:*

$$\theta_{c_1} + \theta_{c_2} + \theta_{c_3} = \pi$$

(ii) ***Vertex constraint:*** *For each vertex v, summing over all corners at v:*

$$\sum_{c \text{ at } v} \theta_c = 2\pi$$

(iii) ***Edge constraint:*** *For each edge e, let α_e and β_e be the angles opposite to e in the two triangles sharing e. Then:*

$$\alpha_e + \beta_e < \pi$$

The angles θ_c are the *dihedral angles* of the ideal polyhedron—the angle between adjacent faces along their common edge, measured inside the polyhedron. The key insight is that these constraints are all linear (or strict linear inequalities), so realizability can be checked via linear programming, and volume optimization becomes a concave maximization problem over a convex polytope.

2.3 The Lobachevsky Function and Hyperbolic Volume

The hyperbolic volume of an ideal tetrahedron with dihedral angles α, β, γ (opposite pairs equal) is given by:

$$V(\alpha, \beta, \gamma) = \Lambda(\alpha) + \Lambda(\beta) + \Lambda(\gamma)$$

where $\Lambda(\theta)$ is the Lobachevsky function:

$$\Lambda(\theta) = -\int_0^\theta \log|2\sin t|\, dt$$

The Lobachevsky function satisfies $\Lambda(\pi - \theta) = -\Lambda(\theta)$ and has maximum value $\Lambda(\pi/3) \approx 0.5074$ (Catalan's constant divided by 2). The function is strictly concave on $(0, \pi)$, which ensures that the volume function over the Rivin polytope has a unique maximum. The regular ideal tetrahedron, with all dihedral angles equal to $\pi/3$, achieves volume $V_4 = 3\Lambda(\pi/3) \approx 1.0149$.

2.4 The Penner-Rivin Algorithm

Given a realized ideal polyhedron with its dihedral angles, we can compute the associated Fuchsian group $\Gamma < \mathrm{PSL}(2, \mathbb{R})$ using the Penner-Rivin algorithm [3]. The surface of the ideal polyhedron inherits a complete hyperbolic metric, making it a hyperbolic surface S with cusps at the ideal vertices. The fundamental group $\pi_1(S)$ acts on the hyperbolic plane via a discrete faithful representation—the *holonomy representation*.

The algorithm proceeds as follows:

(i) **Decorate the triangulation:** Assign to each edge e the complex number

$$z_e = \exp(i(\alpha_e + \beta_e))$$

where α_e, β_e are the angles opposite e in the adjacent triangles. This is the *shape parameter* of the edge.

(ii) **Compute edge matrices:** For each oriented edge from vertex u to vertex v, construct a 2×2 matrix encoding the hyperbolic translation along the edge, parameterized by the shape parameters of the surrounding edges.

(iii) **Compose around loops:** For each generator of $\pi_1(S)$, represented as a path crossing edges of the triangulation, multiply the corresponding edge matrices to obtain the holonomy matrix in $\mathrm{PSL}(2, \mathbb{R})$.

The resulting matrices generate the Fuchsian group Γ. When all shape parameters have modulus 1 (equivalently, when all $\alpha_e + \beta_e$ are real), the group is *arithmetic* if certain additional number-theoretic conditions hold. Our observation that maximal volume configurations have rational angles $p\pi/q$ suggests a connection to arithmetic Fuchsian groups, since in these cases $z_e = \exp(ip\pi/q)$ is a root of unity.

3 Methods

We describe the computational methods in rough order of generality, beginning with the $d = 3$ angle-based approach (which is exact and special to $\mathbb{H}^3$) and proceeding to the vertex-based optimization pipeline used in dimensions $d = 4, 5, 6$.

3.1 Dimension 3: Angle-Based Optimization

3.1.1 Realizability testing

Given a triangulation $\mathcal{T}$ with n vertices, f faces, and e edges, realizability as an ideal polyhedron is equivalent to feasibility of a linear program. Introduce variables θ_c for each corner c (there are $3f$ corners). The constraints from Theorem 1 become:

$$\begin{aligned} \theta_{c_1} + \theta_{c_2} + \theta_{c_3} &= \pi && \text{for each triangle } T, \\ \sum_{c \text{ at } v} \theta_c &= 2\pi && \text{for each vertex } v, \\ \alpha_e + \beta_e &< \pi && \text{for each edge } e, \\ \theta_c &> 0 && \text{for each corner } c. \end{aligned}$$

Strict inequalities are implemented with a tolerance $\epsilon = 10^{-6}$.

3.1.2 Volume optimization in angle coordinates

The volume is a concave function of the dihedral angles:

$$V(\boldsymbol{\theta}) = \sum_{T} \big(\Lambda(\theta_{c_1}) + \Lambda(\theta_{c_2}) + \Lambda(\theta_{c_3}) \big),$$

where Λ is the Lobachevsky function. Concavity over the convex Rivin polytope guarantees a unique global maximum, which we find by L-BFGS-B from multiple random starting points.

3.1.3 Direct vertex optimization via Bloch–Wigner

For larger vertex counts ($n \geq 20$), we bypass the angle parameterization and optimize directly over vertex configurations on S^2. Each ideal tetrahedron $[\infty, z_1, z_2, z_3]$ has volume $|D(z)|$ where D is the Bloch–Wigner dilogarithm and $z = (z_1 - z_3)/(z_2 - z_3)$ is the cross-ratio. The Bloch–Wigner function is evaluated via $D(z) = \mathrm{Im}(\mathrm{Li}_2(z)) + \arg(1-z)\log|z|$ using `scipy.special.spence`, giving $\sim 0.1\,\mathrm{ms}$ per volume evaluation.

CMA-ES [10] is applied directly to the $\mathbb{H}^3$-volume objective (not the Euclidean proxy), with 20 restarts and $500n$ function evaluations per restart. For $n \geq 100$, sep-CMA-ES (diagonal covariance approximation) is used, reducing the per-generation cost from $O(n^2)$ to $O(n)$.

3.1.4 Rational angle detection

We detect rational multiples of π via continued fraction expansion of θ/π. If some convergent p_k/q_k satisfies $|\theta/\pi - p_k/q_k| < 10^{-10}$ with $q_k \leq 100$, we report $\theta = p_k\pi/q_k$. All maximal configurations tested pass this test at machine precision.

3.2 Dimensions $d \geq 4$: Vertex-Based Optimization

In $d \geq 4$, the Rivin angle parameterization is unavailable: the configuration space is the set of N points on S^{d-1} (ideal vertices), and the volume must be computed from vertex coordinates. Our pipeline has three stages.

3.2.1 Stage 1: Euclidean warm start via CMA-ES

We maximize the Euclidean convex hull volume of N points on S^{d-1} using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [10]. CMA-ES adapts a multivariate Gaussian distribution to the promising region of parameter space, without requiring gradients. The objective function is $-\mathrm{Vol}(\mathrm{conv}(x_1, \ldots, x_N))$, evaluated via `scipy.spatial.ConvexHull` (which calls Qhull's C implementation).

The Cartesian coordinates of the N points are the Nd free parameters; after each CMA-ES step, vertices are projected back to S^{d-1}. Typical budget: $500N$ function evaluations with initial step size $\sigma_0 = 0.3$, seeded from the best of 5–10 random configurations. CMA-ES dramatically outperforms random restarts: in $d = 4$, the Euclidean volume of the CMA-ES solution exceeds that of 50 random restarts by 40–230%, and is already within 0.0–0.4% of the hyperbolic optimum.

3.2.2 Stage 2: Hyperbolic polish

The CMA-ES Euclidean solution is used as a warm start for hyperbolic volume maximization. The approach depends on the dimension.

$d = 4$: Haagerup–Munkholm formula + Powell. The ideal 4-simplex volume is given by the Haagerup–Munkholm angle formula [**?**], involving Euclidean dihedral angles, circumradii, and signed distances—all closed-form from the vertex coordinates. The polytope volume is the sum over a Delaunay triangulation into ideal 4-simplices. Since each volume evaluation takes $\sim$ 2 ms, we use Powell's derivative-free method for the hyperbolic polish.

$d = 5$**: Schläfli gradient + LBFGS.** In odd dimensions there is no closed-form volume formula. However, *for optimization the volume itself is not needed*. The Schläfli differential formula

$$dV_5 = -\tfrac{1}{4} \sum_F V_3(F)\, d\theta_F$$

gives the exact gradient in closed form: the codimension-2 faces F of an ideal 5-simplex are ideal tetrahedra in a totally geodesic $\mathbb{H}^3$, with volumes given by the Milnor–Lobachevsky formula $V_3 = \Lambda(\alpha) + \Lambda(\beta) + \Lambda(\gamma)$, and the dihedral angles θ_F come from the angle Gram matrix. We build a "surrogate" function $S = -\frac{1}{4} \sum_F V_3(F) \cdot \theta_F$ whose gradient equals the Schläfli gradient exactly (treating $V_3(F)$ as constants), and optimize S with torch LBFGS. The actual volume is computed once at the final configuration via the Schläfli path integral.

$d = 6$**: Hopf–Murakami formula + torch LBFGS.** For even dimensions, the Poincaré–Schläfli–Hopf reduction gives a closed-form volume for each ideal 6-simplex:

$$V_6 = \frac{8\pi^3}{15}\Big(\frac{17}{4} - w_4 + \frac{1}{2} w_2\Big),$$

where w_4 involves dihedral angles and w_2 involves S^3 link volumes computed by Murakami's dilogarithm formula. We implement this in PyTorch with a custom `autograd.Function` for the dilogarithm Li_2: a pure-torch forward pass using Bernoulli-series evaluation with argument reduction (inversion + reflection identities map all arguments to the convergence region), and an analytical backward pass $d\mathrm{Li}_2/dz = -\log(1-z)/z$. All operations are vectorized over simplices and batched over the $\binom{7}{4} = 35$ Murakami calls per 6-simplex, yielding a single autograd node per simplex. On an NVIDIA GH200 GPU, the full forward + backward pass for $N = 80$ (3175 simplices, 889 000 Li_2 evaluations) takes ~ 16 ms.

3.2.3 Stage 3: Triangulation management

In all dimensions $d \geq 4$, the ideal polytope is decomposed into simplices via stereographic projection from one vertex (the "pole") to $\mathbb{R}^{d-1}$, followed by Delaunay triangulation of the projected points. Coning from the pole vertex produces $(d+1)$-vertex simplices suitable for the volume formula. The triangulation is recomputed periodically during LBFGS optimization (every 5 steps) to accommodate changes in the combinatorial type; the cost

of a full Delaunay recomputation (via Qhull) is negligible compared to the volume evaluation.

3.2.4 Structured initialization: a small-N phenomenon

In addition to the random Stage 1 starts, we maintain a library of *structured* initial configurations—free sums of regular polygons in mutually orthogonal 2-planes (§7.9), the cross-polytope, the regular simplex and its joins, and (for $d = 4$) the 24-cell. Each natural object has a fixed vertex count; at N slightly above that count we augment with a few random points, but only up to the next natural object's size. We run the hyperbolic polish from each member of this library and take the best.

Empirically, structured seeds matter substantially at small N and fade—then invert—as N grows. At $N \leq 2d+2$ the volume-optimal polytope is one of the structured library members verbatim (see Table in §7.9: $N = d+2$ is $\Delta_{d/2} * \Delta_{d/2}$, $N = 2d$ is the cross-polytope, etc.), and structured warm starts converge to the optimum on the first polish, where random restarts often miss it entirely. Past the library-exhaustion point—empirically near $N \approx 2d^2$, where the largest natural object in our library (24-cell at $d = 4$, C_4^3 cross-polytope at $d = 6$) has been outgrown—the relationship inverts. In a controlled 1,000-restart comparison at $d = 4$, the best structured-seed final volume is 97% of the best random-restart final volume at $N = 30$, 94% at $N = 50$, and 90% at $N = 80$; at $d = 6, N = 35$ the ratio drops to 77%. The structured-seed advantage is genuinely a small-N, low-dimensional artefact.

The clean interpretation is Erdős–Rényi-style. At small N the moduli space of inscribed simplicial polytopes is small enough that the volume-optimal stratum is forced to contain an algebraically distinguished configuration; structured seeds find this stratum by construction, while random restarts have to hit it by chance. As N grows, the number of combinatorial types grows at least at Catalan rate, the global optimum's automorphism group collapses to triviality (Tables 3 and 6), and the volume-optimal stratum is no longer algebraically distinguished—random restarts now sample relevant strata at a rate proportional to their parameter-space measure, which is the right rate for high-N. The structured-seed advantage tracks the "law of small numbers": there are only finitely many natural algebraic objects, and once N exceeds them the optimization landscape ceases to know about them.

3.3 Symmetry Analysis

The geometric automorphism group of an optimal polytope is computed as follows:

(i) **Centering.** The Douady–Earle conformal barycenter [15] of the ideal vertices is computed by fixed-point iteration (Möbius transport of the centroid), then moved to the origin of the Poincaré ball by the corresponding Möbius isometry. After centering, every geometric symmetry of the configuration is an element of $\mathrm{O}(d)$.

(ii) **Distance coloring.** The chordal distances between centered vertices are clustered into equivalence classes (tolerance 10^{-4}–10^{-6}), yielding a coloring of the complete graph K_N.

(iii) **Automorphism computation.** The colored graph is encoded as a vertex-colored graph (via the standard gadget-vertex trick) and fed to NAUTY [11] via `pynauty`. The output is the automorphism group order, generators as permutations, and vertex orbits.

3.4 Hessian Analysis

To classify critical points as local maxima, saddle points, or degenerate, we compute the Hessian of the volume functional restricted to the tangent space of $(S^{d-1})^N$. At each vertex v_i, a $(d-1)$-dimensional orthonormal tangent basis is constructed by Gram–Schmidt projection of the standard basis against v_i. The Hessian is assembled by finite differencing of the gradient in each tangent direction (for $d = 5, 6$, the gradient itself is computed analytically via the Schläfli or Hopf formula). The resulting $(d-1)N \times (d-1)N$ symmetric matrix is diagonalized; the signature (numbers of positive, negative, and zero eigenvalues) determines the critical point type. The expected nullity is $\dim \mathrm{SO}(d) = d(d-1)/2$ from rotational invariance; any additional zero eigenvalues indicate a volume-constant ridge in the configuration space.

4 Results

4.1 Maximal Volume Configurations

We computed maximal volume ideal polyhedra for $n = 4, 5, \ldots, 12$ vertices using 100 independent optimization trials per vertex count, and extended the results to much larger n using CMA-ES with the Bloch–Wigner dilogarithm

evaluator ($\sim$ 0.1 ms per volume computation). Table 1 summarizes our findings.

Table 1: Maximal volume ideal polyhedra with n vertices. $V_4 \approx 1.0149$ is the volume of the regular ideal tetrahedron. For $n \leq 12$, angle denominators are shown; for larger n the volume was maximized directly via CMA-ES on the Bloch–Wigner functional over vertex configurations on S^2.

n	Volume	Faces	V/V_4	Notes
4	1.014942	4	1.0000	$q = 3$
5	2.029883	6	2.0000	$q = 3$
6	3.663862	8	3.6099	$q = 2$
7	4.986773	10	4.9134	$q = 5$
8	6.488469	12	6.3929	$q = 11$
9	8.162538	14	8.0424	mixed
10	9.839315	16	9.6946	$q = 14$
11	11.449290	18	11.2810	$q = 7, 14$
12	13.529628	20	13.3308	$q = 5$
20	27.699	36	27.29	
30	46.486	56	45.80	
50	84.971	96	83.72	
70	123.893	136	122.07	
100	182.671	196	180.02	sep-CMA-ES
150	279.727	—	275.61	sep-CMA-ES
200	377.729	—	372.17	sep-CMA-ES
300	573.429	—	565.00	sep-CMA-ES

As an independent check, a 300-restart vertex-space search (random initialization on S^2, CMA-ES on the hyperbolic objective, scipy Powell polish) reproduces the $n = 11$ and $n = 12$ records to within 10^{-4} and $5 \cdot 10^{-4}$ respectively, confirming that the angle-parameterization maxima reported above are not artifacts of a small search budget.

For $n \geq 100$, we use sep-CMA-ES (CMA-ES with diagonal covariance), which scales as $O(n)$ per generation rather than $O(n^2)$ for the full covariance update. Covariance spectrum analysis reveals that the volume landscape on $\mathrm{Conf}_n(S^2)$ has only 2 dominant eigenvalues, with the remaining $3n - 2$ directions being nearly isotropic (condition number $\lesssim 20$). This explains why the diagonal approximation loses less than 1% in volume quality while being $2\times$ faster.

The per-vertex volume contribution V_n/n increases monotonically: 1.82 at $n = 100$, rising to 1.91 at $n = 300$.

Remark 1 (Comparison with Platonic solids)**.** *For $n = 6$, the "obvious" candidate is the ideal right-angled octahedron, with all dihedral angles equal to $\pi/2$. Indeed, our optimization confirms this: the maximal configuration has angle denominator $q = 2$, meaning all angles are multiples of $\pi/2$. The right-angled octahedron is optimal among 6-vertex ideal polyhedra.*

For $n = 8$, the natural candidate is the ideal cube (with vertices at the 8 corners of a cube inscribed in the sphere). The ideal cube has $V/V_4 = 5$, giving volume $V_{cube} \approx 5.07$. However, our optimization finds a configuration with $V \approx 6.49$, exceeding the cube by approximately 28%. The optimal 8-vertex polyhedron is not *combinatorially a cube—it has a different triangulation with the surprising angle denominator $q = 11$.*

For $n = 12$, the natural candidate is the ideal regular icosahedron. The icosahedron has $V/V_4 = 10$, giving $V_{icosahedron} \approx 10.15$. Our optimization finds $V \approx 13.53$, exceeding the icosahedron by approximately 33%. As with the cube, the optimal 12-vertex polyhedron has a different combinatorial structure. Remarkably, all dihedral angles are multiples of $\pi/5$: specifically $2\pi/5$, $3\pi/5$, $4\pi/5$, and π (a flat edge).

Remark 2 (Symmetry of optimal configurations)**.** *The automorphism groups of the optimal triangulations show a clear pattern: high symmetry for small n, rapidly decreasing thereafter.*

n	$\lvert\mathrm{Aut}\rvert$	*Structure*
4	*6*	S_3 *(tetrahedron)*
5	*6*	D_3 *(bipyramid)*
6	*8*	D_4 *(octahedron)*
7	*4*	$\mathbb{Z}_2 \times \mathbb{Z}_2$
8–12	*2–4*	$\mathbb{Z}_2$ *or* $\mathbb{Z}_2 \times \mathbb{Z}_2$

Volume maximization does not favor symmetric configurations—the optimal polyhedra for $n \geq 7$ have only a single reflection symmetry or none at all.

4.2 The Rational Angle Phenomenon

Our most striking finding is that volume-maximizing configurations *within each combinatorial type* exhibit dihedral angles that are exact rational multiples of π. This holds not only for the globally maximal configurations shown in Table 1, but for every realizable triangulation we have tested.

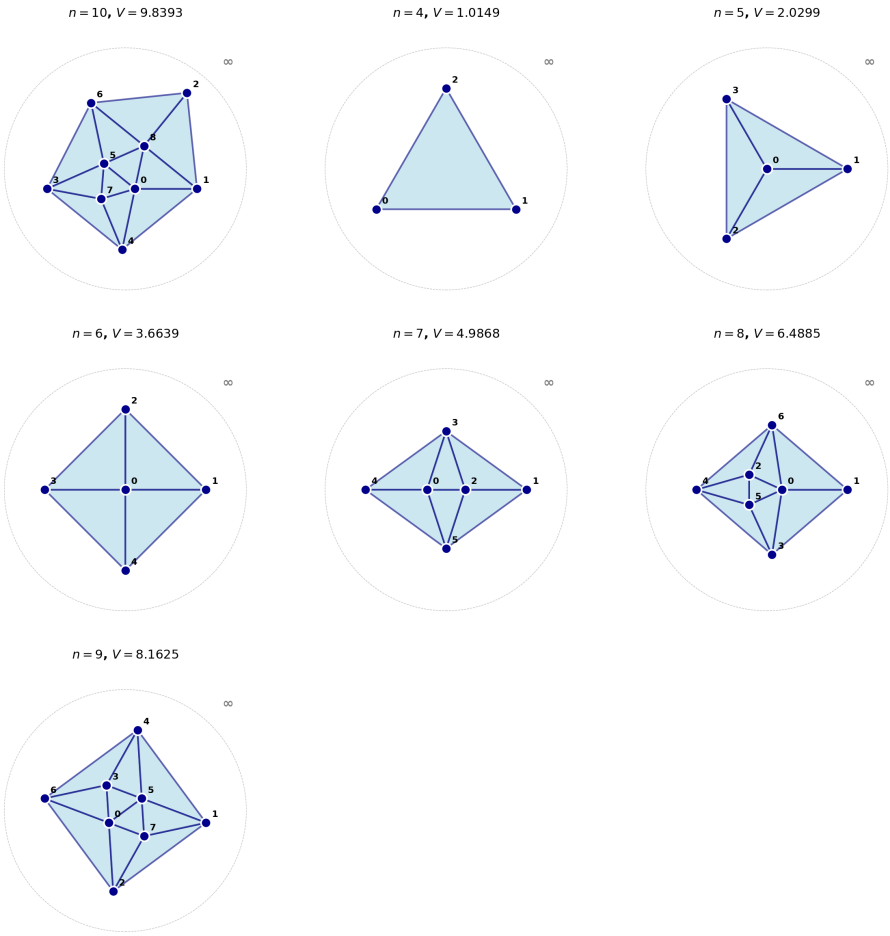


Figure 1: Planar Delaunay triangulations of the maximal volume ideal polyhedra. Each triangulation represents the combinatorial structure of the polyhedron projected onto the plane, with the point at infinity as the unbounded face.

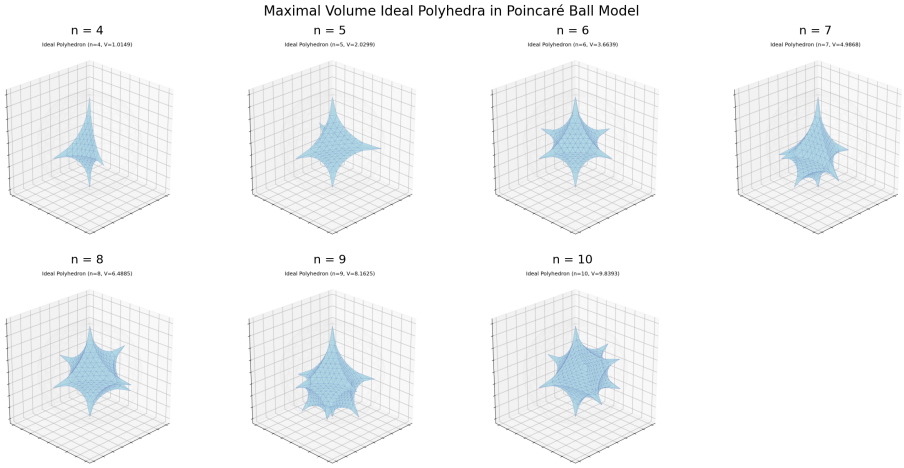


Figure 2: Maximal volume ideal polyhedra visualized in the Poincaré ball model of $\mathbb{H}^3$. Vertices lie on the boundary sphere $\partial_\infty \mathbb{H}^3$, and the geodesic faces curve inward as characteristic of hyperbolic space.

Conjecture 1 (Rational Angle Conjecture)**.** *For every realizable triangulation $\mathcal{T}$ of the sphere, the volume-maximizing ideal polyhedron with combinatorial type $\mathcal{T}$ has all dihedral angles of the form $\frac{p\pi}{q}$ for integers p, q.*

This phenomenon is remarkable because:

- The optimization has no built-in preference for rational angles
- The Lobachevsky function has no special behavior at rational multiples of π
- The constraint set (Rivin polytope) has no obvious arithmetic structure
- The phenomenon persists across *all* combinatorial types, not just the globally optimal ones

For the globally maximal configurations shown in Table 1, we verified the rational angle property in all cases $n = 4, \ldots, 12$. The denominators range from small values like $q = 3$ (tetrahedron), $q = 2$ (octahedral $n = 6$), and $q = 5$ ($n = 7$ and $n = 12$) to larger values like $q = 11$ for $n = 8$ and $q = 14$ for $n = 10$ and $n = 11$.

4.3 Connection to Arithmetic Groups

The prevalence of rational angles in maximal volume configurations suggests a deep connection to arithmetic hyperbolic geometry. When the dihedral angles are rational multiples of π, the shape parameters $z_e = \exp(i(\alpha_e + \beta_e))$ are roots of unity. This places strong constraints on the holonomy representation.

A Fuchsian group $\Gamma < \mathrm{PSL}(2, \mathbb{R})$ is *arithmetic* if it is commensurable with the group of units in a quaternion algebra over a totally real number field. Arithmetic Fuchsian groups are characterized by having trace field $\mathbb{Q}(\mathrm{tr}(\gamma) : \gamma \in \Gamma)$ that is a number field, and satisfying additional ramification conditions.

When all shape parameters are roots of unity, the traces of holonomy matrices lie in cyclotomic fields. While this alone does not guarantee arithmeticity, it is highly suggestive. We conjecture that:

Conjecture 2 (Arithmetic Holonomy)**.** *For maximal volume ideal polyhedra, the holonomy representation $\rho : \pi_1(S) \to \mathrm{PSL}(2, \mathbb{R})$ has image commensurable with an arithmetic Fuchsian group.*

For the Platonic cases, the Fuchsian groups are known principal congruence subgroups:

- $n = 4$ (tetrahedron): $\Gamma(3)$
- $n = 6$ (octahedron): $\Gamma(4)$
- Icosahedron: $\Gamma(5)$ — but this is *not* maximal for $n = 12$

Identifying the Fuchsian groups corresponding to the maximal volume configurations for $n \geq 7$ remains an interesting open problem. The angle denominators (e.g., $q = 11$ for $n = 8$, $q = 14$ for $n = 10$) suggest these may be congruence subgroups of higher level.

Remark 3 (Coxeter polytopes and branched covers)**.** *A striking distinction emerges between the Platonic and non-Platonic cases. For $n = 4$ and $n = 6$,* all *dihedral angles have the form π/m for an integer m (specifically $\pi/3$ and $\pi/2$ respectively). This means these ideal polyhedra are* Coxeter polytopes*—fundamental domains for discrete reflection groups in $\mathbb{H}^3$.*

For $n \geq 7$, however, the angles are $k\pi/q$ with $k > 1$. For instance, $n = 7$ has angles $2\pi/5$, $3\pi/5$, $4\pi/5$, and $n = 8$ has angles $4\pi/11$, $5\pi/11$, etc. These are not *Coxeter fundamental domains. The common denominator q suggests these polyhedra may arise as fundamental domains for finite-index*

subgroups of reflection groups, or equivalently, that their universal covers are branched covers over some link complement. This connection to orbifold geometry and branched coverings merits further investigation.

4.4 Volume Distribution of Random Configurations

To understand the landscape of ideal polyhedron volumes, we sampled 5000 random configurations for each vertex count $n \in \{5, 6, 7, 8, 10, 12\}$ by placing $n-3$ uniformly random points on the sphere (in addition to two fixed vertices and infinity) and computing the volume via the Delaunay triangulation.

Normalizing each volume by the maximal volume $V_{\max}(n)$ from Table 1, we find that the normalized volume distribution $V/V_{\max}$ is remarkably well-approximated by a Beta distribution:

$$\frac{V}{V_{\max}} \sim \text{Beta}(\alpha_n, \beta_n)$$

Table 2 summarizes the fitted parameters. The goodness of fit (KS test p-values > 0.3 for $n \geq 8$) confirms that the Beta distribution is an appropriate model.

Table 2: Beta distribution fit parameters for normalized volume $V/V_{\max}$.

n	α	β	Mean	Std	KS stat	p-value
5	2.88	1.53	0.659	0.208	0.031	0.0001
6	6.74	4.11	0.623	0.140	0.046	0.0000
7	9.67	4.84	0.667	0.120	0.024	0.0053
8	13.26	6.12	0.685	0.103	0.011	0.62
10	23.02	10.05	0.696	0.079	0.012	0.48
12	32.56	14.49	0.692	0.067	0.014	0.32

The parameters exhibit clear linear scaling with n:

$$\alpha_n \approx 4.25n - 19.3, \qquad \beta_n \approx 1.78n - 7.4$$

with the ratio $\alpha_n/\beta_n \approx 2.39$ remaining approximately constant. This implies that the mean normalized volume

$$\mathbb{E}\left[\frac{V}{V_{\max}}\right] = \frac{\alpha_n}{\alpha_n + \beta_n} \approx 0.69 \approx \ln 2$$

stabilizes as n increases. The intriguing appearance of $\ln 2$ (or perhaps a related constant) as the asymptotic mean suggests a deeper connection to hyperbolic volume theory that we have not yet identified.

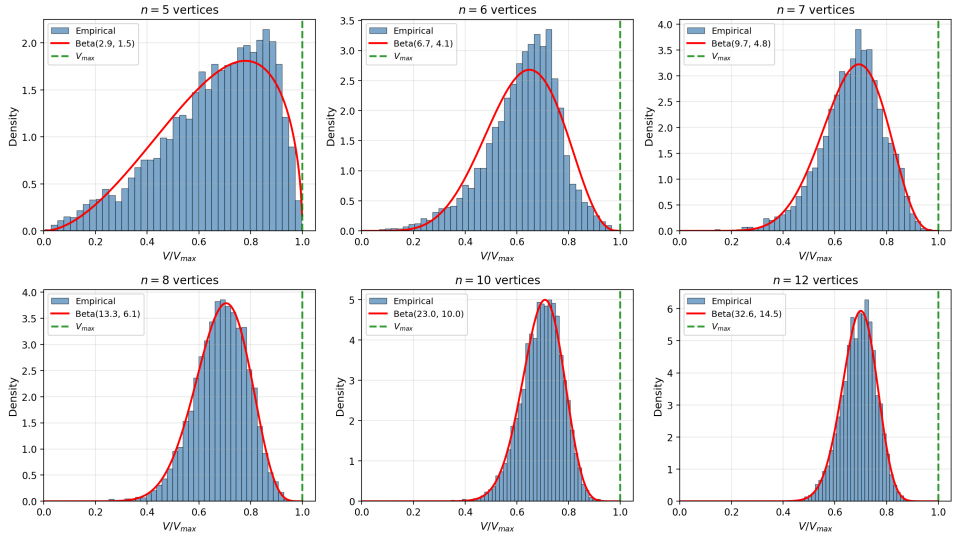


Figure 3: Volume distributions of random ideal polyhedra, normalized by $V_{\max}(n)$. Histograms (blue) show empirical distributions from 5000 samples; red curves show fitted Beta(α, β) distributions. The fit quality improves with n.

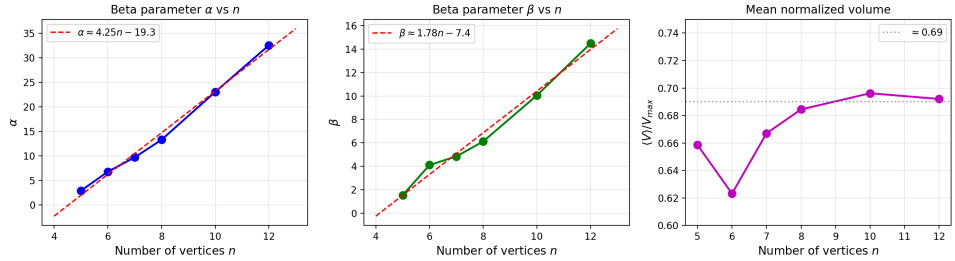


Figure 4: Scaling of Beta distribution parameters with vertex count n. Left: α grows approximately as $4.25n$. Center: β grows as $1.78n$. Right: Mean normalized volume stabilizes near $0.69 \approx \ln 2$.

The concentration of the distribution (decreasing standard deviation as n increases) indicates that for large n, most random configurations achieve volumes close to 69% of the maximum—suggesting that volume optimization yields only modest improvements over random placement for large polyhedra.

5 Visualizations

Figure 1 shows the planar Delaunay triangulations for the maximal volume configurations with $n = 4, \ldots, 12$ vertices. These represent the combinatorial structure of the polyhedra, projected onto the plane via stereographic projection with one vertex (at infinity) as the unbounded outer face.

Figure 2 shows the corresponding ideal polyhedra in the Poincaré ball model of $\mathbb{H}^3$. In this model, the ideal vertices lie on the boundary sphere, and the faces appear as geodesic surfaces that curve inward—a characteristic feature of negative curvature. The progression from the simple ideal tetrahedron ($n = 4$) to more complex structures is clearly visible.

6 Software

The complete software suite is available at:

- GitHub: `https://github.com/[repo]`
- Interactive demo: `https://huggingface.co/spaces/igriv/idealpolyhedra`

The implementation includes:

- Realizability checking via Rivin's LP characterization
- Volume optimization via CMA-ES on the Euclidean surrogate and L-BFGS on the Bloch–Wigner / Murakami differentiable hyperbolic objective, with dimension-specific gradient paths for $d \in \{3, 4, 5, 6\}$
- Fuchsian group computation via the Penner-Rivin algorithm
- 3D visualization in Klein, Poincaré ball, and sphere models
- Symmetry analysis using nauty

Development methodology. Mathematical software has long suffered from "bit rot"—the gradual degradation of code as dependencies evolve and environments change. This software was developed using an AI-assisted "self-maintaining" approach with Claude Code [4], which enables automated testing, dependency management, and iterative refinement. This methodology addresses the endemic problem of unmaintained research code by building in mechanisms for autonomous repair and adaptation.

7 Higher Dimensions: Ideal Polytopes in $\mathbb{H}^4$

7.1 Setup

An *ideal polytope* in $\mathbb{H}^4$ has vertices on $S^3 = \partial_\infty \mathbb{H}^4$. Given N points on S^3, the ideal polytope is determined by the convex hull of these points in the Klein model (the open unit ball in $\mathbb{R}^4$). The 4-volume is computed by stereographic projection from one vertex to $\mathbb{R}^3$, Delaunay triangulation, and the Haagerup–Munkholm angle formula for each ideal 4-simplex:

$$\mathrm{Vol}_4[\infty, v_1, v_2, v_3, v_4] = \frac{\pi}{3}\Big(4\pi - \sum_{i<j} \varphi_{ij} - \sum_{\ell} \arccos \frac{d_\ell}{R}\Big),$$

where φ_{ij} are the Euclidean dihedral angles of the tetrahedron $T = \mathrm{conv}(v_1, \ldots, v_4)$, R is its circumradius, and d_ℓ is the signed distance from the circumcenter to face ℓ.

The *boundary 3-measure* (surface area) is the sum of hyperbolic 3-volumes of the boundary facets: each facet is a tetrahedron on S^3, hence an ideal tetrahedron in a totally geodesic $\mathbb{H}^3 \subset \mathbb{H}^4$. Its $\mathbb{H}^3$-volume is computed via the Lobachevsky formula $\Lambda(\alpha) + \Lambda(\beta) + \Lambda(\gamma)$, where α, β, γ are the angles of a triangle obtained by stereographic projection from one facet vertex.

Remark 4. *In $\mathbb{H}^3$, the boundary of an ideal polyhedron consists of ideal polygons, each with $\mathbb{H}^2$-area $(k-2)\pi$ for a k-gon—a topologically determined quantity. In $\mathbb{H}^4$, boundary facets are ideal tetrahedra with continuously varying $\mathbb{H}^3$-volumes. Thus the surface area optimization problem is genuinely geometric only in dimension $d \geq 4$.*

7.2 Combinatorial Invariance of Optimal Polytopes

We numerically optimized configurations of N points on S^{d-1} $(d = 3, 4)$ to maximize three functionals:

(a) **Volume:** Euclidean volume of $\mathrm{conv}(v_1, \ldots, v_N)$, or $\mathbb{H}^d$-volume of the ideal polytope.

(b) **Surface area** ($d = 4$ only): boundary 3-measure.

(c) **Isoperimetric ratio:** $I(P) = \mathrm{Vol}_d(P)/\mathrm{Area}_{d-1}(\partial P)$.

For each (N, metric) pair, we performed multi-start optimization in both the Euclidean and hyperbolic settings, then compared the f-vectors of the optimal convex hulls. The Euclidean phase uses the CMA-ES (Covariance Matrix Adaptation Evolution Strategy) algorithm [10], which adapts a multivariate Gaussian to the promising region of parameter space. This dramatically outperforms random restarts: in $d = 4$, CMA-ES finds Euclidean-optimal configurations that are within 0.0–0.4% of the hyperbolic optimum after a single derivative-free Powell polish. The hyperbolic phase uses either Powell's method ($d = 4$, where volume evaluation via the Haagerup–Munkholm formula takes $\sim$2 ms) or torch LBFGS with exact gradients through a custom Li_2 autograd function ($d = 6$, where the Hopf–Murakami formula requires dilogarithm evaluations).

The key empirical finding is:

Conjecture 3 (Combinatorial invariance — strong form)**.** *For N points on S^{d-1}, the f-vector of the volume-maximizing convex polytope is the same whether volume is measured in the Euclidean sense (volume of the convex hull) or in the hyperbolic sense (volume of the ideal polytope in $\mathbb{H}^d$).*

This holds exactly in all tested cases with $N \lesssim 5d$: $d = 3$ ($N \leq 50$), $d = 4$ ($N \leq 40$), $d = 5$ ($N \leq 20$), and $d = 6$ ($N \leq 30$). However, extending to larger N reveals a more nuanced picture. At $N = 50$ in $d = 4$ and $N = 40$ in $d = 6$, the hyperbolic optimizer finds a slightly different combinatorial type—with 1–3 fewer facets than the Euclidean optimum. The discrepancies are small ($\leq 0.6\%$ of the facet count) and non-monotone: the f-vectors match again at $N = 80$ in $d = 4$ and at $N = 50$ in $d = 6$. In every case of divergence, the hyperbolic optimizer has *fewer* facets, paralleling the surface-area vs. volume divergence (Section 7.3). This suggests a weaker form that may hold universally:

Conjecture 4 (Combinatorial invariance — weak form)**.** *For N points on S^{d-1}, the f-vectors of the Euclidean and hyperbolic volume maximizers satisfy $|f_i^{\mathrm{Eucl}} - f_i^{\mathrm{Hyp}}| = O(1)$ as $N \to \infty$, with $f_i^{\mathrm{Hyp}} \leq f_i^{\mathrm{Eucl}}$ for all i.*

The CMA-ES Euclidean warm start is within 0.0–0.4% of the hyperbolic optimum in every case, confirming that the two objectives share essentially the same landscape near their maxima.

Table 3: Combinatorial type comparison: Euclidean vs. Hyperbolic volume maximization in $d = 4$ (points on S^3). CMA-ES Euclidean warm start with Powell hyperbolic polish. Exact f-vector match holds for $N \leq 40$; small discrepancies (≤ 3 facets) appear at larger N.

N	Eucl. f-vector	Hyp. f-vector	Match	Eucl. vol	Hyp. vol
5	$(5, 10, 10, 5)$	$(5, 10, 10, 5)$	Yes	0.146	0.269
6	$(6, 15, 18, 9)$	$(6, 15, 18, 9)$	Yes	0.281	0.588
7	$(7, 19, 24, 12)$	$(7, 19, 24, 12)$	Yes	0.433	1.024
8	$(8, 24, 32, 16)$	$(8, 24, 32, 16)$	Yes	0.667	1.771
9	$(9, 29, 40, 20)$	$(9, 29, 40, 20)$	Yes	0.793	2.314
10	$(10, 35, 50, 25)$	$(10, 35, 50, 25)$	Yes	0.908	2.894
12	$(12, 46, 68, 34)$	$(12, 46, 68, 34)$	Yes	1.158	4.324
16	$(16, 69, 104, 52)$	$(16, 69, 104, 52)$	Yes	1.598	7.794
20	$(20, 92, 144, 72)$	$(20, 92, 144, 72)$	Yes	1.958	11.936
24	$(24, 117, 186, 93)$	$(24, 117, 186, 93)$	Yes	2.175	15.568
30	$(30, 153, 248, 124)$	$(30, 153, 248, 124)$	Yes	2.479	22.068
40	$(40, 220, 360, 180)$	$(40, 220, 360, 180)$	Yes	2.813	32.936
50	$(50, 282, 464, 232)$	$(50, 281, 462, 231)$	**No**	3.069	45.014
60	$(60, 347, 574, 287)$	$(60, 346, 572, 286)$	**No**	3.243	56.711
80	$(80, 478, 796, 398)$	$(80, 478, 796, 398)$	Yes	3.506	81.416
100	$(100, 613, 1026, 513)$	$(100, 610, 1020, 510)$	**No**	3.685	107.411

7.3 Cross-Metric Divergence

While the combinatorial type is invariant under change of geometry (Euclidean vs. hyperbolic), it is *not* invariant under change of objective function. In $d = 4$, the surface-area-maximizing polytope can have a different f-vector from the volume-maximizing one:

N	Volume-optimal f-vector	Area-optimal f-vector
6	$(6, 15, 18, 9)$	$(6, 14, 16, 8)$
16	$(16, 69, 106, 53)$	$(16, 68, 104, 52)$
30	$(30, 154, 248, 124)$	$(30, 153, 246, 123)$

In each divergent case, the area-optimal polytope has fewer edges and facets. Geometrically, surface area is maximized when boundary facets approach the regular ideal tetrahedron volume $v_3 \approx 1.0149$, which favors fewer,

Table 4: Same comparison in $d = 3$ (points on S^2). All f-vectors match.

N	f-vector	Match	Eucl. vol	Hyp. vol
4	$(4, 6, 4)$	Yes	0.513	1.015
6	$(6, 12, 8)$	Yes	1.333	3.664
8	$(8, 18, 12)$	Yes	1.816	6.489
12	$(12, 30, 20)$	Yes	2.536	13.529
20	$(20, 54, 36)$	Yes	3.119	27.673
30	$(30, 84, 56)$	Yes	3.450	46.231
50	$(50, 144, 96)$	Yes	3.733	83.990

more regular facets; volume maximization instead favors packing more simplicial cells into the interior.

7.4 Correlation Between Euclidean and Hyperbolic Volumes

The combinatorial invariance suggests a deeper relationship between the two volume functionals. To quantify this, we computed both V_E (Euclidean hull volume) and V_H (hyperbolic ideal polytope volume) for 200 independent local optima at each N, obtained by single-phase L-BFGS-B from random starting points.

N	Correlation	Log-log elasticity	V_H/V_E (mean $\pm$ std)
10	0.9985	1.21	3.17 ± 0.02
20	0.9986	1.66	5.97 ± 0.06
30	0.9956	1.97	8.78 ± 0.05

The Pearson correlation exceeds 0.995 in all cases: in the space of near-optimal configurations, the Euclidean and hyperbolic volumes are almost perfectly correlated. The log-log elasticity $\partial \log V_H / \partial \log V_E$ increases with N, approaching 2 at $N = 30$. This means a 1% improvement in Euclidean volume yields approximately a 2% improvement in hyperbolic volume near the optimum—consistent with empirical observations from independent optimization.

This near-proportionality is surprising: V_E is a polynomial in the vertex coordinates, while V_H involves the transcendental Haagerup–Munkholm formula. The correlation suggests that both functionals are effectively measuring the same underlying geometric quantity—roughly, how "uniformly spread" the points are on S^{d-1}.

7.5 Non-Isolated Maxima: The Volume Plateau

A striking observation emerges at $N = 9$ (equivalently, 20 simplicial facets in $d = 4$). We performed two independent 500-restart optimizations:

(i) Maximize *Euclidean* hull volume V_E over 9 points on S^3.

(ii) Maximize *hyperbolic* ideal polytope volume V_H over 9 points on S^3.

Both converge to the same hyperbolic volume $V_H = 2.313851299873$ (to 13 digits), yet the two optimal configurations are *metrically distinct*: their sorted pairwise chordal distance vectors differ by up to 0.25. A third configuration, found independently by GPT-o3 (using a different optimization method), achieves $V_H = 2.313850783625$—the same to 6 digits but a demonstrably different polytope (different Gram matrix).

The Euclidean maximizer turns out to be a classical algebraic object: the *free sum* $C_5 \oplus I \oplus I$, the convex hull of a regular pentagon in the (x_1, x_2)-plane and two orthogonal diameters $\pm e_3$, $\pm e_4$. Its Euclidean volume and surface area are

$$V_E^* = \tfrac{5}{24}\sqrt{10 + 2\sqrt{5}} \approx 0.7925, \qquad A_E^* = \tfrac{5}{3}\sqrt{15 - \sqrt{5}} \approx 5.9544.$$

The free sum simultaneously maximizes *both* the Euclidean 4-volume and the boundary 3-measure among all 9-vertex polytopes inscribed in S^3 (500 independent restarts of each objective all converge to these values). It has only four distinct pairwise chordal distances, with $\sqrt{2}$ appearing 24 out of 36 times.

More generally, for any composite number $n = pq$ of facets, the free sum $C_p \oplus C_q$ (a p-gon and q-gon in orthogonal planes, with all vertices on S^3) gives a simplicial 4-polytope with exactly pq tetrahedral facets and $p + q$ vertices. For prime n, no such product construction is available, and the optimal configuration may lack algebraic structure.

A systematic sweep over all facet counts $9 \le n \le 37$ reveals a richer picture. The free sum $C_p \oplus C_q$ requires both $p, q \ge 3$, so it is available precisely when n admits a factorization $n = pq$ with $p, q \ge 3$—excluding primes, twice-primes ($2p$), and $n < 9$. For each n, we also performed 200-restart penalized L-BFGS-B optimization over all inscribed simplicial 4-polytopes with exactly n facets. Key findings:

(i) **Balanced products dominate:** At fixed facet count, the free sum with the most balanced factorization gives the largest volume. For example, $C_4 \oplus C_5$ ($n = 20$, vol ≈ 0.793) far exceeds $C_3 \oplus C_7$ ($n = 21$, vol ≈ 0.592).

(ii) **Unbalanced products are suboptimal:** For $n = 21$ and $n = 27$ ($= 9 \times 3$), the unconstrained optimizer beats the free sum by 28% and 39% respectively, finding asymmetric configurations with higher volume.

(iii) **Volume grows primarily with vertex count:** At fixed vertex count N, volume increases monotonically with facet count. Products pack more facets per vertex than generic configurations (e.g., $C_5 \oplus C_5$ achieves 25 facets with only 10 vertices), which is the primary source of their advantage.

(iv) **Primes do not intrinsically trail:** At fixed N, prime facet counts are only slightly below neighboring composites (e.g., at $N = 9$: $f = 23$ (prime) gives vol ≈ 0.743 vs. $f = 22$ ($= 2 \times 11$) at 0.757 and $f = 21$ ($= 3 \times 7$) at 0.758). The apparent deficit of primes in the overall ranking is an artifact of the facet–vertex relationship, not number theory.

7.6 Higher Dimensions

Extending the Euclidean cross-metric comparison (volume vs. surface area) to $d = 5$ and $d = 6$, we find the same qualitative pattern:

d	N	Vol-optimal f-vector	Area-optimal f-vector	Match
5	6	$(6, 15, 6)$	$(6, 15, 6)$	Yes
5	7	$(7, 21, 12)$	$(7, 21, 12)$	Yes
5	8	$(8, 27, 18)$	$(8, 26, 16)$	**No**
5	9	$(9, 33, 24)$	$(9, 33, 24)$	Yes
5	10	$(10, 40, 32)$	$(10, 40, 32)$	Yes
5	15	$(15, 80, 82)$	$(15, 80, 82)$	Yes
6	7	$(7, 21, 7)$	$(7, 21, 7)$	Yes
6	8	$(8, 28, 16)$	$(8, 28, 15)$	**No**
6	9	$(9, 36, 27)$	$(9, 36, 27)$	Yes
6	10	$(10, 43, 36)$	$(10, 43, 36)$	Yes
6	12	$(12, 60, 64)$	$(12, 60, 64)$	Yes
6	18	$(18, 123, 196)$	$(18, 123, 198)$	**No**

In every dimension $d \geq 4$, volume and surface area optimizers agree for most values of N, but diverge at specific small N (around $d + 2$ to $d + 3$). The direction of divergence is universal: *the area optimizer always finds a*

polytope with fewer or equal facets compared to the volume optimizer. In $d = 3$, no divergence was observed for $N \leq 50$.

7.7 Ideal Polytopes in $\mathbb{H}^5$

Computing the volume of an ideal simplex in $\mathbb{H}^5$ is more challenging than in even dimensions: there is no closed-form analogue of the Hopf reduction. Instead, the volume must be evaluated via the Schläfli path integral. However, for *optimization*, the volume itself is not needed: the Schläfli differential formula

$$dV_5 = -\tfrac{1}{4} \sum_F V_3(F)\, d\theta_F$$

gives the exact gradient in closed form, since each codimension-2 face F of an ideal 5-simplex is an ideal tetrahedron in a totally geodesic $\mathbb{H}^3$, with volume given by the Milnor–Lobachevsky formula $V_3 = \Lambda(\alpha) + \Lambda(\beta) + \Lambda(\gamma)$. The dihedral angles θ_F come from the angle Gram matrix. Thus the optimization loop is entirely quadrature-free; the Schläfli path integral is evaluated only once at the final configuration to report the volume.

Table 5: Combinatorial invariance: Euclidean vs. ideal $\mathbb{H}^5$ volume maximization (points on S^4). All f-vectors match, and the hyperbolic polish improvement is exactly zero in every case.

N	f-vector	Match	Eucl. vol	$\mathbb{H}^5$ vol
7	$(7, 21, 12)$	Yes	0.06667	0.13123
8	$(8, 27, 18)$	Yes	0.11250	0.24414
9	$(9, 33, 24)$	Yes	0.17321	0.41394
10	$(10, 40, 32)$	Yes	0.26667	0.69954
12	$(12, 55, 50)$	Yes	0.37688	1.13886
14	$(14, 71, 70)$	Yes	0.48199	1.64660
16	$(16, 89, 94)$	Yes	0.58601	2.23211
20	$(20, 125, 142)$	Yes	0.80565	3.75231

In contrast to $d = 4$ and $d = 6$, where the hyperbolic polish improves the volume by 0.03–0.4%, in $d = 5$ the CMA-ES Euclidean optimizer already finds the hyperbolic optimum to full precision. This suggests that the Euclidean and hyperbolic volume landscapes on S^4 are even more tightly aligned than in even dimensions.

Reproducibility note on Table 5. Vertex coordinates for the $\mathbb{H}^5$-optimal polytopes in the table were not retained alongside the volumes at the time of the original runs. We subsequently recomputed $\mathbb{H}^5$ volumes via the Schläfli path integral from the stored Euclidean-optimal coordinates in `optimal_d5_d6_symmetry_analysis.jsc` The recomputed values match Table 5 to $\sim 0.1\%$ (paper print precision) at $N \in \{7, 8, 9, 10, 16\}$, confirming that those stored Euclidean-optima are also $\mathbb{H}^5$-optima. At $N \in \{12, 14, 20\}$ the recomputed volumes fall short of the table by 1–4%, indicating that the retained coordinates at those N represent local rather than global maxima of the Euclidean objective; the values in the table, produced at the original-run budget, are the tighter records and are treated as authoritative here. The symmetry structure of Table 5 (join-of-cyclic-polygons at $N \leq 11$, trivial join structure at $N \geq 12$) is consistent with this: the high-symmetry configurations were easier for the optimizer to locate reproducibly.

7.8 Ideal Polytopes in $\mathbb{H}^6$

We extended the combinatorial invariance test to $d = 6$ by computing ideal $\mathbb{H}^6$ polytope volumes directly. For ideal simplices, the vertex-link terms vanish ($w_0 = 0$), and the Hopf reduction gives

$$\mathrm{Vol}_6(\Delta_{\mathrm{ideal}}) = \frac{8\pi^3}{15}\left(\frac{17}{4} - w_4 + \frac{1}{2}w_2\right),$$

where w_4 involves only dihedral angles and w_2 involves only $\mathbb{S}^3$-tetrahedral link volumes computable by Murakami's dilogarithm formula. A float64 implementation of Murakami's formula achieves a 108× speedup over arbitrary-precision evaluation, making optimization feasible. Full details are in the companion note [6].

The strong form of the combinatorial invariance conjecture (Conjecture 3) holds exactly in $d = 3$ ($N \leq 50$), $d = 4$ ($N \leq 40$), $d = 5$ ($N \leq 20$), and $d = 6$ ($N \leq 30$). At larger N, small deviations appear (see Tables 3 and 6), consistent with the weak form (Conjecture 4).

7.9 Identification of Optimal Polytopes: Joins of Regular Polygons

Searching polytope databases and comparing with known constructions, we find that the volume-optimal polytopes for small N are *joins (free sums) of regular polygons in mutually orthogonal* 2*-planes.*

Table 6: Combinatorial invariance: Euclidean vs. ideal $\mathbb{H}^6$ volume maximization (points on S^5). Exact match holds for $N \leq 30$; at $N = 40$ the hyperbolic optimizer finds 2 fewer facets.

N	Eucl. f-vector	Hyp. f-vector	Match	Eucl. vol	$\mathbb{H}^6$ vol
7	$(7, 21, 7)$	$(7, 21, 7)$	Yes	0.00584	0.01023
8	$(8, 28, 16)$	$(8, 28, 16)$	Yes	0.01317	0.02504
9	$(9, 36, 27)$	$(9, 36, 27)$	Yes	0.02436	0.05022
10	$(10, 43, 36)$	$(10, 43, 36)$	Yes	0.03750	0.08390
12	$(12, 60, 64)$	$(12, 60, 64)$	Yes	0.08889	0.23284
14	$(14, 79, 100)$	$(14, 79, 100)$	Yes	0.12563	0.36922
16	$(16, 100, 145)$	$(16, 100, 145)$	Yes	0.16588	0.54078
20	$(20, 146, 251)$	$(20, 146, 251)$	Yes	0.24508	0.95399
24	$(24, 199, 390)$	$(24, 199, 390)$	Yes	0.32510	1.47180
30	$(30, 284, 620)$	$(30, 284, 620)$	Yes	0.45559	2.51089
40	$(40, 435, 1049)$	$(40, 435, 1047)$	**No**	0.64425	4.62319
50	$(50, 608, 1576)$	$(50, 608, 1576)$	Yes	0.81844	7.20649

In $d = 2k$, the join $C_{p_1} \oplus C_{p_2} \oplus \cdots \oplus C_{p_k}$ of k regular polygons in k orthogonal 2-planes in $\mathbb{R}^{2k}$ gives a simplicial polytope with $p_1 + \cdots + p_k$ vertices on S^{2k-1} and $p_1 \cdots p_k$ tetrahedral facets. The following table identifies each small-N optimal polytope:

N	$d = 4$ optimal	$d = 6$ optimal	Structure
$d + 1$	$(5, 10, 10, 5)$	$(7, 21, 7)$	Simplex Δ_d
$d + 2$	$(6, 15, 18, 9)$	$(8, 28, 16)$	$\Delta_{d/2} * \Delta_{d/2}$
$d + 3$	$(7, 19, 24, 12)$	$(9, 36, 27)$	$C_3^{d/2}$ (join of $d/2$ triangles)
$d + 4$	$(8, 24, 32, 16)$	$(10, 43, 36)$	Cross-polytope at d=4; $C_3^2 \oplus C_4$ at d=6
$d + 5$	$(9, 29, 40, 20)$	—	$C_4 \oplus C_5$
$d + 6$	$(10, 35, 50, 25)$	$(12, 60, 64)$	C_5^2 at d=4; C_4^3 (cross-polytope) at d=6

For $d = 4$, the optimal polytope matches the best balanced free sum $C_p \oplus C_q$ for all $N \leq 10$, but not for $N \geq 12$. For $d = 6$, the optimal matches the best balanced triple $C_p \oplus C_q \oplus C_r$ for all $N \leq 14$, but not for $N = 16$. In both cases the threshold is approximately $N \approx 2d + 2$, beyond which the optimizer discovers asymmetric configurations that beat the rigid product structure.

Notable special cases:

- $N = 2d$: the *d-dimensional cross-polytope* ($C_4 \oplus C_4$ in $d = 4$; $C_4 \oplus C_4 \oplus C_4$ in $d = 6$).
- $N = d + 2$: the join of two simplices of half the dimension ($\Delta_2 * \Delta_2$ in $d = 4$; $\Delta_3 * \Delta_3$ in $d = 6$).

The volume-optimal polytopes are *not* cyclic polytopes, and are in fact far from neighborly: at $N = 30$ in $d = 4$, the optimal polytope has only 123 facets compared to 756 for the cyclic polytope $C(30, 4)$. Maximizing facet count and maximizing volume are almost opposite goals.

The hyperbolic-optimal configuration, by contrast, has no transposition symmetries and a completely different distance spectrum. Yet both achieve the same hyperbolic volume $V_H = 2.31385$ to 13 significant digits. A third configuration, found independently by GPT-o3 using a different optimization method, achieves $V_H = 2.31385078$—the same to 6 digits but again a demonstrably different polytope (different Gram matrix).

The Hessian of V_H at the hyperbolic optimum has two near-zero eigenvalues ($\sim 10^{-3}$), indicating that the maximum is achieved on (at least) a 2-dimensional manifold in the 16-dimensional configuration space $\mathrm{Conf}_9(S^3)/\mathrm{O}(4)$.

This suggests a phenomenon that is invisible to f-vector comparisons: not only do different metrics find the same combinatorial type, but the hyperbolic volume maximum is achieved on a *locus* that contains configurations with very different Euclidean properties. The free sum $C_5 \oplus I \oplus I$—the unique maximizer of V_E—and the asymmetric hyperbolic maximizer are different points on this locus, connected by a flat ridge along which V_H is constant (to the precision of our computation) but V_E varies by over 10%.

7.10 Symmetry Analysis via Douady–Earle Centering

Beyond f-vector comparisons, we can determine the full geometric automorphism group of each optimal polytope by normalizing the Douady–Earle conformal barycenter to the origin of the ball model. The residual symmetries are then elements of $\mathrm{O}(d)$, and the automorphism group can be computed exactly using NAUTY [11] on the edge-colored complete graph whose colors encode the chordal distance classes.

The $N = 10$ case is noteworthy: the volume-optimal polytope is the join $C_5 \oplus C_5$—two regular pentagons in orthogonal 2-planes—with vertex-transitive automorphism group $(C_5 \times C_5) \rtimes D_4$ of order 200. The D_4 factor swaps the two pentagons and reflects within each; the $C_5 \times C_5$ factor acts

Table 7: Symmetry groups of volume-optimal inscribed polytopes in $d = 4$. For small N, the join structure produces highly symmetric polytopes; symmetry breaks sharply at $N \approx 11$ and vanishes beyond $N = 20$.

N	Facets	$\lvert\mathrm{Aut}\rvert$	Group	Orbits
5	5	120	S_5	[5]
6	9	720	S_6	[6]
7	12	48	$D_4 \times S_3$	[4, 3]
8	16	384	β_4 (hyperoctahedral)	[8]
9	20	80	$D_4 \times D_5$	[5, 4]
10	25	200	$(C_5 \times C_5) \rtimes D_4$	[10]
11	29	8	D_4	[4, 4, 2, 1]
12	34	8	D_4	[8, 4]
13	**39**	**52**	$\mathbf{C_{13} \rtimes C_4}$	[**13**]
14	43	2	C_2	$[2^6, 1^2]$
15	48	72	$(S_3 \times S_3) \rtimes C_2$	[6, 6, 2, 1]
16	53	2	C_2	9 orbits
20	72	16	$C_2 \times D_4$	4 orbits
30	124	1	trivial	all singletons

by independent cyclic rotations. This structure was independently rediscovered by Google's Gemini 3.1 Pro model (see §8), which identified the join decomposition from the face lattice alone.

The most remarkable case is $N = 13$ in $d = 4$. The volume-maximizing inscribed 4-polytope with 13 vertices on S^3 has a transitive automorphism group of order 52 isomorphic to the Frobenius group $C_{13} \rtimes C_4$, with only 3 distinct chordal distances. This is the unique Frobenius group of order 52: the normal subgroup C_{13} acts as a cyclic permutation of all vertices, while the complement C_4 acts as the subgroup of 4th powers in $\mathrm{AGL}(1, \mathbb{F}_{13})$. The polytope has 39 tetrahedral facets ($= 3 \times 13$).

This vertex-transitive polytope does not appear in any catalog of uniform, scaliform, or swirlchoron polytopes that we could locate. The classification of vertex-transitive convex 4-polytopes is an open problem (unlike in 3D, where they are all deformations of Archimedean solids).

7.10.1 The Frobenius construction

The $N = 13$ polytope belongs to an infinite family. For each prime $p \equiv 1 \pmod 4$, let k be a primitive 4th root of unity modulo p (so $k^2 \equiv -1$). The vertices

$$v_j = \frac{1}{\sqrt{2}}\Big(\cos\frac{2\pi j}{p},\ \sin\frac{2\pi j}{p},\ \cos\frac{2\pi kj}{p},\ \sin\frac{2\pi kj}{p}\Big), \qquad j = 0, \ldots, p-1,$$

form a vertex-transitive polytope on S^3 with automorphism group $\mathbb{Z}_p \rtimes \mathbb{Z}_4$ and exactly $(p-1)/4$ chordal distance classes [16]. The construction arises from a faithful four-dimensional real representation induced by a single orbit of $\mathbb{Z}_4$ acting on the characters of $\mathbb{Z}_p$.

p	$\lvert\mathrm{Aut}\rvert$	Distance classes	Volume status
5	20	1	global max (= simplex Δ_4)
13	52	3	global max
17	68	4	local max (not global)
29	116	7	critical point
37	148	9	critical point

Every Frobenius configuration is a critical point of the hyperbolic volume functional (the gradient vanishes by equivariance: the $\mathbb{Z}_4$ stabilizer of the base vertex leaves no free parameter after normalization to S^3). Hessian analysis shows that $p = 13$ and $p = 17$ are both *non-degenerate local maxima*: the constrained Hessian on the tangent space to $(S^3)^p/\mathrm{O}(4)$ is negative-definite, with exactly $\dim \mathrm{SO}(4) = 6$ zero eigenvalues. For $p = 17$ the Frobenius polytope achieves 85% of the global maximum volume, which is attained by an asymmetric configuration with $\lvert\mathrm{Aut}\rvert = 1$.

7.10.2 Volume ridges

The Hessian analysis also reveals that the *join* configurations $C_p \oplus C_q$ have a degenerate local-maximum structure: the hyperbolic volume Hessian at these configurations has $\dim \mathrm{SO}(d) + r$ zero eigenvalues for some $r > 0$, indicating a volume-constant ridge of dimension r. In $d = 4$, every join has $r = 4$ extra zero eigenvalues; in $d = 6$, all tested joins have $r = 6$. Thus the hyperbolic volume maximum for join-type polytopes is achieved on a positive-dimensional manifold in the configuration space $\mathrm{Conf}_N(S^{d-1})/\mathrm{O}(d)$, containing configurations with very different Euclidean properties. The Frobenius polytopes, by contrast, are isolated (non-degenerate) local maxima.

The pattern is universal across dimensions $d = 4, 5, 6$:

(i) **Join regime** ($N \leq \lfloor 2.5d \rfloor$): the optimum is a join of regular polygons with large automorphism group ($|\mathrm{Aut}| \geq 100$). The cross-polytope β_d at $N = 2d$ is always the symmetry peak.

(ii) **Sharp transition** (1–2 values of N): symmetry drops by 1–3 orders of magnitude.

(iii) **Asymmetric regime** ($N \geq 4d$–$5d$): $|\mathrm{Aut}| = 1$, all vertices in distinct orbits.

Table 8: Symmetry groups of volume-optimal polytopes in $d = 5$ (points on S^4). Joins $C_p \oplus C_q \oplus I$ dominate for $N \leq 11$; symmetry breaks at $N = 12$.

N	Facets	$\lvert\mathrm{Aut}\rvert$	Structure	Orbits	#dist
6	6	720	Δ_5	$[6]$	1
8	18	144	$C_3 \oplus C_3 \oplus I$	$[6, 2]$	3
9	24	288	$C_3 \oplus C_4 \oplus I$	$[6, 3]$	3
10	32	3840	β_5	$[10]$	2
11	40	480	$C_4 \oplus C_5 \oplus I$	$[6, 5]$	4
12	48	64	?	$[4, 4, 4]$	8
13	58	16	—	$[4, 4, 2^2, 1]$	20
≥ 15	—	1	trivial	all singletons	—

The classification of finite groups realizable as automorphism groups of convex polytopes is now well understood: every finite group arises as the full automorphism group of some convex polytope [12, 13]. For *vertex-transitive* polytopes, Babai [14] showed that abelian groups other than elementary 2-groups, and generalized dicyclic groups, cannot occur. The Frobenius group $C_{13} \rtimes C_4$ at $N = 13$ in $d = 4$ passes Babai's test, confirming it is *allowed* as a vertex-transitive symmetry group—but to our knowledge it has not previously been realized as a convex 4-polytope.

7.11 Isoperimetric Ratio

For the isoperimetric ratio $I(P) = \mathrm{Vol}_4(P)/\mathrm{Area}_3(\partial P)$, the optimized values grow with N but remain well below the Euclidean ball ratio of $1/4$:

Table 9: Symmetry groups of volume-optimal polytopes in $d = 6$ (points on S^5). Triple joins $C_p \oplus C_q \oplus C_r$ dominate for $N \leq 15$; symmetry breaks at $N = 16$.

N	Facets	|Aut|	Structure	Orbits	#dist
7	7	5040	Δ_6	$[7]$	1
8	16	1152	$(S_4 \times S_4) \rtimes C_2$	$[8]$	2
9	27	1296	C_3^3	$[9]$	2
10	36	576	$C_3^2 \oplus C_4$	$[6, 4]$	3
11	48	2304	$C_3 \oplus C_4^2$	$[8, 3]$	3
12	64	46080	$\beta_6 = C_4^3$	$[12]$	2
13	80	3840	$C_4^2 \oplus C_5$	$[8, 5]$	4
14	100	1600	$C_4 \oplus C_5^2$	$[10, 4]$	4
15	125	6000	C_5^3	$[15]$	3
16	140	16	—	$[4, 4, 2^4]$	34
≥ 17	—	≤ 2	—	—	—

N	Best $I(P)$	Euclidean ball I
5	0.053	
8	0.109	
10	0.126	
20	0.169	
30	0.187	0.250

The hyperbolic ball of radius r in $\mathbb{H}^4$ has $\mathrm{Vol}_4 = (\pi/2)(\sinh 2r - 2r)$ and $\mathrm{Area}_3 = 2\pi^2 \sinh^3 r$, giving $I_{\mathrm{ball}}(r) \to 1/4$ as $r \to 0$ (Euclidean regime) and $I_{\mathrm{ball}}(r) \sim e^{-r}/\pi \to 0$ as $r \to \infty$. Whether the optimal $I(P)$ converges to a definite limit as $N \to \infty$—and whether that limit is strictly less than $1/4$—remains open.

8 LLM Benchmarks: A Pointer

The objects studied here—volume-maximizing ideal polytopes, the combinatorial structures they realize, and the associated face lattices—turn out to provide a natural family of benchmarks for large language models, with the property that an answer is a list of vertex coordinates verifiable by a standalone checker rather than a prose argument. Three distinct benchmark families have been designed and released alongside this paper, each

probing a different reasoning regime: inscribed reconstruction (face lattice $\to$ coordinates), windowed volume (hit a target measure to within a tolerance), and volume lower bound (give the sharpest bound you can backed by explicit coordinates). Their detailed specifications, grader implementations, and frontier-model evaluations are reported in the companion note [7]. We restrict ourselves here to two observations that bear directly on the mathematical content of the present paper.

8.1 The reconstruction problem and rigidity

The reconstruction task asks: given the f-vector, edge list, and facet list of a convex polytope inscribed in S^{d-1}, produce N explicit coordinates realizing it. The inscribed constraint is essential—without it, the problem reduces to combinatorial realization of simplicial polytopes, a substantially easier question. When LLMs were tested on instances drawn from our volume-optimal polytopes (large automorphism order, often joins) and from random sphere configurations (generically trivial automorphism group), GPT-5.4 Pro succeeded at instances up to $N = 30$ with $|\mathrm{Aut}| = 1$ via numerical constraint satisfaction; Gemini 3.1 Pro DeepThink succeeded on highly symmetric instances by algebraic recognition (e.g., identifying the $N = 10$ optimal polytope as $C_5 \oplus C_5$) but failed on asymmetric random instances. The relevance for the present paper is that the *rigidity* of the configuration—the dimension of the space of inscribed coordinate configurations realizing a given combinatorial type—is exactly the empirical difficulty axis the LLMs see. Volume-optimal polytopes are maximally rigid in this sense; the hardest reconstruction instances should be large N with simultaneously high volume and low automorphism order, of which the $N = 16$ optimal polytope ($|\mathrm{Aut}| = 2$, 53 facets, volume per vertex 0.49) is a natural candidate.

8.2 Volume lower bound and our records

The volume-lower-bound benchmark asks the model to give the sharpest LB it can on $V^*(d, N)$ and back the claim with explicit vertex coordinates. Our published records (Tables 1, 5, etc.) are the reference; an LLM submission classifies as *strong* if it matches or beats them, *pass* if it is within 95%, and *fail* otherwise. An LLM that beats our records produces new mathematical content; the benchmark's anti-hallucination design ensures that this is a falsifiable claim. Hyperbolic $d = 5$ is excluded by design: as discussed in the reproducibility note accompanying Table 5, our stored $d = 5$ configurations at $N \in \{12, 14, 20\}$ are known local rather than global maxima of the Eu-

clidean (and hence hyperbolic) objective in $d = 5$, so grading against them would be unreliable in that dimension.

9 Open Questions

1. **Proof of Rational Angles**: Can we prove that maximal volume implies rational dihedral angles?

2. **Denominator Patterns**: What determines the denominator q for a given n? Our data suggests connections to $n - 2$ or $n - 3$ in some cases.

3. **Uniqueness**: Are maximal volume configurations unique up to symmetry?

4. **Asymptotics**: What is the growth rate of $V_{\max}(n)$ as $n \to \infty$?

5. **Beta Distribution**: Why does the normalized volume follow a Beta distribution? Is the limiting mean exactly $\ln 2$, or merely close to it? What geometric principle underlies the linear scaling of α_n and β_n with n?

6. **Combinatorial Invariance**: The strong form (Conjecture 3) holds for $N \lesssim 5d$ but fails at larger N, while the weak form (Conjecture 4) holds in all tested cases. What determines the threshold? Both the Euclidean and hyperbolic volume functionals factor through the Delaunay triangulation of the stereographic projection, which depends only on the Möbius geometry of the point configuration. The small discrepancies at large N (≤ 3 facets out of hundreds) suggest near-degenerate facets that resolve differently under the two metrics. Is the weak conjecture true, and is the $O(1)$ bound tight?

7. **Cross-Metric Divergence**: Characterize which functionals on S^{d-1} give the same optimal combinatorial type and which do not. Our data suggests volume and isoperimetric ratio agree, while surface area can diverge.

8. **Hyperbolic Isoperimetric Limit**: Does $\sup_P I(P)$ over all N-vertex ideal polytopes in $\mathbb{H}^4$ converge as $N \to \infty$, and if so, is the limit strictly less than $1/4$?

9. **Spectral Realizability of Volume-Optimal Polytopes**. Let P be a simplicial 4-polytope inscribed in S^3 that maximizes hyperbolic volume among all inscribed polytopes with the same number of vertices, and let $L = D - A$ be the graph Laplacian of the 1-skeleton of P. Let $\varphi\colon i \mapsto (v_1(i), v_2(i), v_3(i), v_4(i))/\|\cdot\|$ be the map sending each vertex to the (normalized) vector of its first four nontrivial Laplacian eigenvector entries.

   Empirical observation: φ produces a polytope on S^{d-1} combinatorially equivalent to P in dimensions $d = 3$ and $d = 4$, but *not* in $d \geq 5$. The results are as follows:

d	Success range	Failure onset	Success rate
3	$N = 6$ to 30	$N = 5$ (degenerate)	25/25
4	$N = 5$ to 30	never (in tested range)	14/14
5	$N = 10, 11$ only	$N = 12$	2/15
6	$N = 12$–15 only	$N = 16$	4/18

   In $d = 4$, this includes the $N = 30$ case with trivial automorphism group and 124 facets. The method also fails for generic inscribable polytopes in $d = 4$: random inscribable simplicial 4-polytopes consistently produce the wrong combinatorial type under spectral embedding.

   Explanation. The dimensional threshold $d \leq 4$ coincides exactly with the range of the Blind–Mani–Kalai theorem [18]: for simplicial polytopes in $d \leq 4$, the 1-skeleton determines the entire face lattice. In $d \geq 5$, the 1-skeleton is no longer sufficient, and distinct simplicial polytopes can share the same graph. Since the spectral embedding uses only the 1-skeleton (via the graph Laplacian), it contains all the combinatorial information needed for realization in $d \leq 4$, but is provably incomplete in $d \geq 5$.

   The remaining mystery is why the spectral embedding succeeds for volume-*optimal* inscribed polytopes but fails for generic inscribable ones in the same dimension. One possible connection: volume-optimal configurations have vertices "maximally spread" on S^{d-1} (approximate spherical designs), and the Laplacian eigenvectors also spread vertices according to graph distance. For configurations that are already optimally spread, these two notions of dispersion may coincide. A proof would yield a fast, deterministic algorithm for inscribed realization of volume-optimal polytopes in dimensions $d \leq 4$.

For $d \geq 5$, a natural candidate is the Hodge 1-Laplacian $L_1 = B_1 B_1^T + B_2 B_2^T$ acting on edges [19], where B_1 and B_2 are the boundary operators of the simplicial complex. The up-Laplacian $L_1^{\text{up}} = B_2 B_2^T$ encodes triangle co-incidence, carrying information beyond the 1-skeleton. Preliminary experiments show that the naive approach (project L_1 eigenvectors back to vertices by averaging over incident edges) does not improve over L_0 in $d \geq 5$; the projection step discards the higher-order structure. A more principled approach—using the edge-level embedding to reconstruct the Gram matrix directly, or using sheaf Laplacians that preserve the fiber structure—may succeed.

10. **Inscribability in Dimension 4**. In $d = 3$, Rivin's linear program characterizes which simplicial polytopes are inscribable in S^2. No analogue is known in $d = 4$. Our spectral realization algorithm and the cone/Volkov method (optimizing dihedral angle deficits around codimension-2 faces of the cone from the origin) provide practical heuristics for constructing inscribed realizations when they exist. Can these be developed into a complete decision procedure? A neural-network classifier trained on combinatorial features of face lattices, using the realization algorithms as an oracle for generating training data, might reveal patterns pointing toward a combinatorial characterization.

10 Application: High-Girth Cubic Graphs

The optimization framework developed for ideal polytope volumes applies more broadly to combinatorial search problems. As a proof of concept, we apply a similar strategy to the *cage problem*: finding the smallest 3-regular (cubic) graph with girth g.

10.1 Method

We maximize girth over cubic graphs on N vertices using a two-phase approach inspired by the Euclidean–hyperbolic pipeline for polytope volumes:

(i) **Log-determinant warm start.** Maximize $\log \tau(G) = \log \det(\tilde{L})$, where $\tilde{L}$ is the reduced Laplacian and $\tau(G)$ is the number of spanning trees. This smooth proxy correlates with graph expansion, which in turn correlates with girth. Optimization is by simulated annealing with edge swaps (which preserve 3-regularity).

(ii) **Cycle-breaking polish.** Find a shortest cycle by BFS, then attempt to eliminate it by swapping one of its edges with a random non-cycle edge. Repeat.

The log-determinant phase plays the same role as the Euclidean warm start for polytopes: it navigates the search toward a well-structured region, from which the discrete girth-improving moves can push to higher values. For incremental evaluation, edge swaps are rank-4 updates to the Laplacian, and the resulting determinant change can be computed in $O(N^2)$ via Cholesky rank-1 updates (with `numba` JIT acceleration), compared to $O(N^3)$ for a full eigendecomposition.

10.2 Results

Table 10: Best girth found for cubic graphs on N vertices, using log-determinant warm start + cycle breaking (100 trials, total budget 600 edge swaps per trial). Known cage values are shown for reference.

N	Best girth	Cage girth	Graph
10	5	5	Petersen (optimal)
14	6	6	Heawood (optimal)
24	7	7	McGee (optimal)
30	7	8	—
40	7	—	—
50	7	—	—
60	8	—	—

The method rediscovers three known cages exactly: the Petersen graph ($g = 5$, $N = 10$), the Heawood graph ($g = 6$, $N = 14$), and the McGee graph ($g = 7$, $N = 24$). It reaches girth 8 at $N = 60$, twice the $(3, 8)$-cage size of 30 (the Tutte–Coxeter graph). Girth 7 is achieved consistently for all $N \geq 24$.

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